Food Delivery SQL Analytics Project

Project By,

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Project Overview:

- This project focuses on analyzing a food delivery dataset using SQL to extract meaningful business insights.
- The goal is to understand customer behavior, order patterns, and growth metrics through real-world analytical queries.
- Table includes,

- Customer code
- Restaurant id
- Cuisine
- Placed at
- Promo code name
- Order status

Objectives:

- Identify top-performing restaurants by cuisine
- Analyze customer acquisition trends over time
- Track user behavior and order frequency
- Measure organic vs promoted customer growth
- Support data-driven decision making.

1. Find Top 2 outlets by cuisine type without using limit and top function.

```sql
with cte as (
select cuisine, restaurant_id, count(*) as no_of_orders
from orders
group by cuisine, restaurant_id)
select * from (select *,
row_number() over (partition by cuisine order by no_of_orders desc) as rn
from cte) a
where rn<=2;
```

Result Grid | Filter Rows: | Export:

cuisine	restaurant_id	no_of_orders	rn
American	BURGER99	8	1
American	AMERICAN2	6	2
Italian	PIZZA123	10	1
Italian	ITALIAN2	6	2
Japanese	SUSHI456	6	1
Japanese	JAPANESE2	5	2
Lebanese	KMKMH6787	10	1
Lebanese	LEBANESE2	9	2
Mexican	TACO789	7	1
Mexican	MEXICAN2	6	2

Insights: Top restaurants dominate order volume within each cuisine, showing strong customer preference.
A few outlets drive most of the demand, indicating uneven performance across partners.

2. Find the daily new customer count from the launch date (everyday how many new customers are we acquiring).

```sql
with cte as (
select customer_code,  cast(min(placed_at)as date) as first_order_date
from orders
group by customer_code)
select first_order_date, count(*) as no_of_new_customers
from cte
group by first_order_date
order by first_order_date;
```

Result Grid | Filter Rows:

first_order_date	no_of_new_customers
2025-01-01	2
2025-01-02	1
2025-01-03	1
2025-01-04	1
2025-01-05	3
2025-01-06	1
2025-01-07	1
2025-01-08	1
2025-01-09	1
2025-01-10	3
2025-01-11	1
2025-01-12	1
2025-01-13	1
2025-01-14	1
2025-01-15	2
2025-01-16	1
2025-01-17	1

Result Grid | Filter Rows:

first_order_date	no_of_new_customers
2025-01-16	1
2025-01-17	1
2025-01-18	1
2025-01-19	1
2025-01-20	2
2025-01-21	1
2025-01-22	1
2025-01-23	1
2025-01-24	1
2025-01-25	1
2025-01-26	1
2025-01-27	1
2025-01-28	1
2025-01-29	1
2025-01-30	1
2025-01-31	4
2025-02-01	2

Insights: Customer acquisition fluctuates daily, with noticeable spikes on certain days.
This suggests marketing campaigns or external factors significantly impact user growth.

3. Count of all the users who were acquired in Jan 2025 and only placed one order in Jan and did not place any other order.

```sql
select customer_code, count(*) as no_of_orders
from orders
where month(placed_at)=1 and year(placed_at)=2025
and customer_code not in (select distinct customer_code
from orders
where month(placed_at) between 2 and 12 and year(placed_at)=2025)
group by customer_code
having count(*)=1
order by customer_code;
```

Insights: A segment of users placed only one order and never returned, indicating early churn.
This highlights the need to improve first-time user experience and retention strategies.

Result Grid | Filter Rows:

customer_code	no_of_orders
BCD7890123456ABC	1
DEF5678901234MNO	1
EFG1234567890DEF	1
FGH7890123456GHI	1
GHI3456789012MNO	1
GHI5678901234XYZ	1
HIJ9876543210DEF	1
IJK1234567890JKL	1
JAN_ONLY_ORDER1	1
JAN_ONLY_ORDER2	1
JKL3456789012XYZ	1
JKL7890123456MNO	1
KLM5678901234DEF	1
LMN9876543210JKL	1
MNO1234567890PQR	1
MNO7890123456XYZ	1
NOP3456789012DEF	1
OPQ5678901234JKL	1
PQR1234567890ABC	1
PQR9876543210PQR	1
QRS7890123456DEF	1

Result Grid | Filter Rows:

customer_code	no_of_orders
KLM5678901234DEF	1
LMN9876543210JKL	1
MNO1234567890PQR	1
MNO7890123456XYZ	1
NOP3456789012DEF	1
OPQ5678901234JKL	1
PQR1234567890ABC	1
PQR9876543210PQR	1
QRS7890123456DEF	1
RST3456789012JKL	1
SINGLE_ORDER_JAN	1
STU5678901234PQR	1
STU9876543210ABC	1
TUV1234567890GHI	1
VWX3456789012PQR	1
VWX5678901234ABC	1
WXY9876543210GHI	1
XYZ1234567890MNO	1
YZA3456789012ABC	1
ZAB5678901234GHI	1

4. Growth team is planning to create a trigger that will target customers after their every third order with a personalized communication and they asked you to create a query for this.

```sql
with cte as (select *,
row_number() over (partition by customer_code order by placed_at) as order_number
from orders
)
select *
from cte
where order_number%3=0
 and date(placed_at) = '2025-03-31';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_id	Customer_code	Placed_at	Restaurant_id	Cuisine	Order_status	Promo_code_Name	order_number
OF1900191864	LAST_ORDER_7DAYS	2025-03-31 16:30:00	KMKMH6787	Lebanese	Delivered	NULL	3
OF1900191870	MULTI_CUISINE_CUST	2025-03-31 14:45:00	PIZZA123	Italian	Delivered	NULL	6

Insights: Customers reaching every third order indicate strong engagement and repeat behavior. These users are ideal for loyalty rewards and personalized marketing to increase retention.

5. List customers who placed more than 1 order and all their orders on a promo only.

```
select customer_code, count(*) as no_of_orders, count(promo_code_name) as promo_orders
from orders
group by customer_code
having count(*)>1 and count(*) = count(promo_code_name);
```

Result Grid | Filter Rows: | Export:

customer_code	no_of_orders	promo_orders
DEF9876543210XYZ	2	2
UVW7890123456JKL	3	3

Insights: Some customers rely entirely on promo codes for all their orders.
This indicates high price sensitivity and risk of churn without discounts.

6. What percent of customers were organically acquired in Jan 2025 (placed their first order without promo code).

```sql
with cte as (
select *,
row_number() over (partition by customer_code order by placed_at) as rn
from orders
where month(placed_at)=1
)
select count(
case
 when rn=1 and promo_code_name is null then customer_code
 end)*100/count(distinct customer_code) as organic_customers_percent
from cte;
```

Result Grid | Filter Rows:

	organic_customers_percent
▶	43.9024

Insights: Around 44% of customers were acquired organically without promo codes.
This shows decent brand strength but also reliance on promotional strategies for growth.

Key Insights:

- Customer acquisition trends vary significantly across dates
- A portion of users rely entirely on promo-based purchases
- Organic customers contribute significantly to long-term growth
- Identifying repeat behavior helps in targeted engagement strategies.

Skills Demonstrated:

- SQL Joins
- Aggregations
- Subqueries
- Windows Functions
- CTEs

Conclusion:

- This project strengthened my ability to analyze data using SQL and derive actionable insights.
- It highlights customer preferences, revenue drivers, and operational patterns, helping businesses make informed decisions.